CS 231 Semester Project

Project Report Component – 2 (Description) (Flow diagram & Screenshots of the project)

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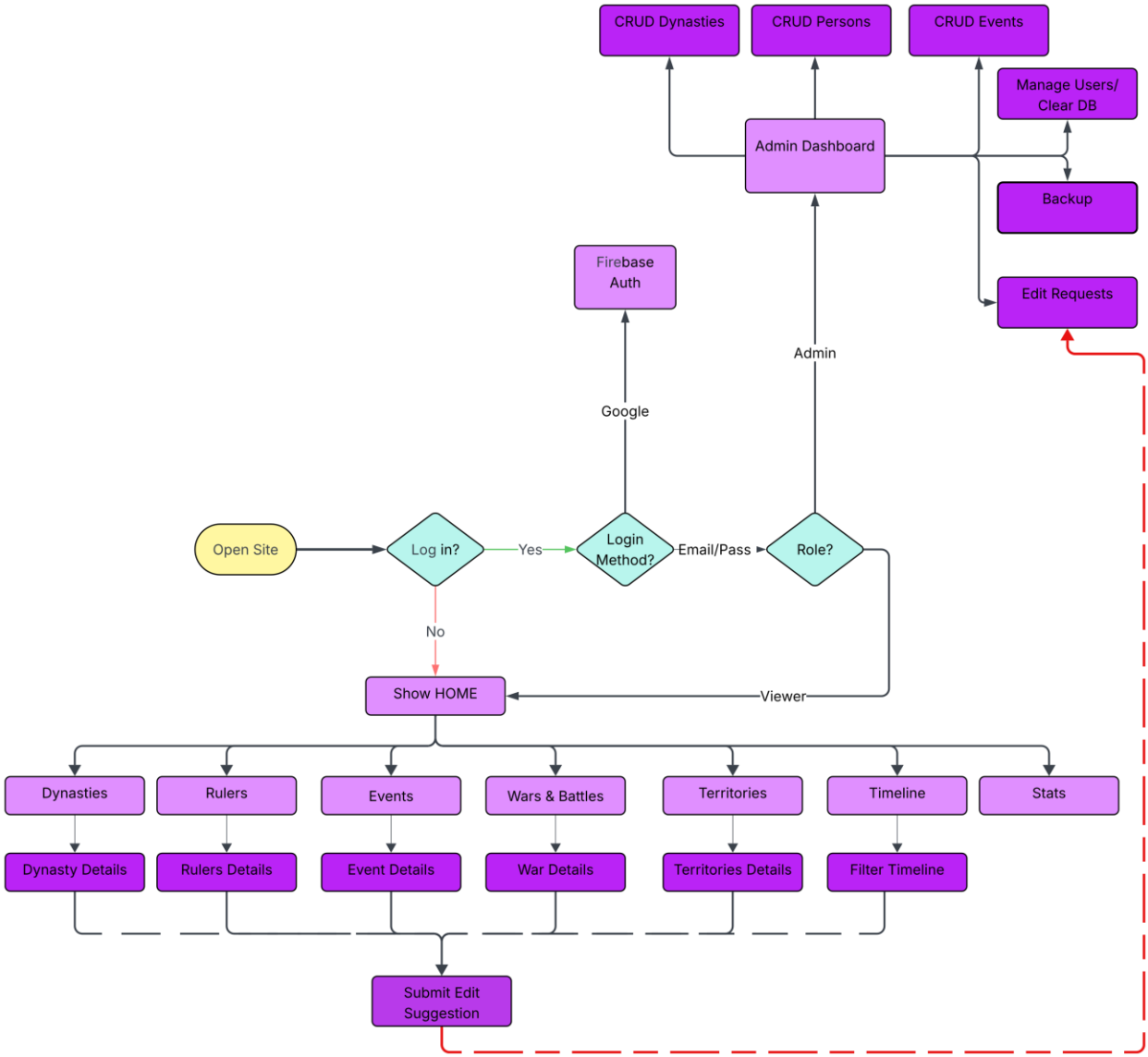
Flow Diagram:



Description:

Authentication

1. **Login Screen** — Enter username and password to log in; also has a Google Sign-In button for Firebase-based authentication.
2. **Register Screen** — Enter username, email, and password to create a new viewer account.

Public / Viewer Screens

3. **Home Screen** — Landing page showing high-level counts (total dynasties, rulers, events, territories) and entry points to all sections.
4. **Dynasties List Screen** — Browse all dynasties with their name, time period, and description. Click any dynasty to see its detail page.
5. **Dynasty Detail Screen** — Shows full information for one dynasty including its rulers, controlled territories, associated events, and succession chain.
6. **Rulers List Screen** — Browse all historical persons with filters for name search, dynasty, title, era (start/end year), and sorting options.
7. **Ruler Detail Screen** — Full profile of one ruler including biography, reign(s), family relations (spouses, children), succession links, and events they participated in. Registered users see a Suggest Edit button.
8. **Events List Screen** — Browse all historical events with an optional filter by event type (battle, war, political, natural disaster, death).
9. **Event Detail Screen** — Full details of one event including date, location, description, outcome, linked dynasty, and persons involved.
10. **Wars & Battles Screen** — Filtered view showing only war and battle type events. Supports filtering by ruler, year range, and type (war only / battle only / both).
11. **War/Battle Detail Screen** — Same as Event Detail but accessed from the Wars page; shows outcome, commanders involved, and related dynasty.

12. **Territories List Screen** — Browse all territories showing region, modern name, and which dynasties controlled them and when.
13. **Territory Detail Screen** — Full profile of one territory including its description, modern equivalent, and a timeline of dynasty control periods.
14. **Timeline Screen** — Visual chronological view of events across history. Filters for year range, dynasty, ruler, and event type.
15. **Stats Screen** — Aggregated insights such as longest reigns, succession chains, number of events per dynasty, and territory control durations.
16. **Search Screen** — Unified search across rulers, dynasties, events, and territories. Returns categorised results from a single query.
17. **Suggest Edit Screen** — Available to logged-in viewers only. Allows submitting a correction or addition for a specific ruler or dynasty field, stored as a pending edit request.

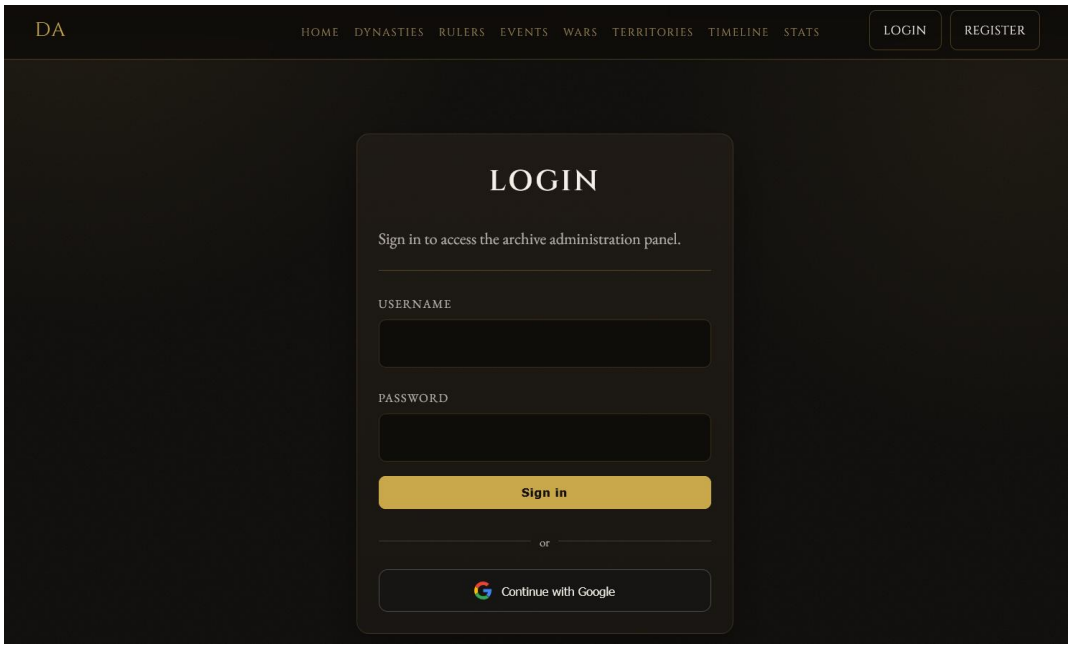
Admin Screens

18. **Admin Dashboard Screen** — Overview for admins showing record counts and quick links to all management sections.
19. **Admin Dynasties List Screen** — Table of all dynasties with edit and delete buttons, plus an Add Dynasty button.
20. **Add / Edit Dynasty Screen** — Form to create or update a dynasty record including name, start/end year, description, and image URL.
21. **Admin Persons List Screen** — Table of all persons with edit and delete buttons, plus an Add Person button.
22. **Add / Edit Person Screen** — Form to create or update a ruler record including name, dates, biography, dynasty, image, and optional first reign details (title, capital, dates). Also manages spouse, parent/child, succession, and event links.
23. **Admin Events List Screen** — Table of all events with edit and delete buttons, plus an Add Event button.

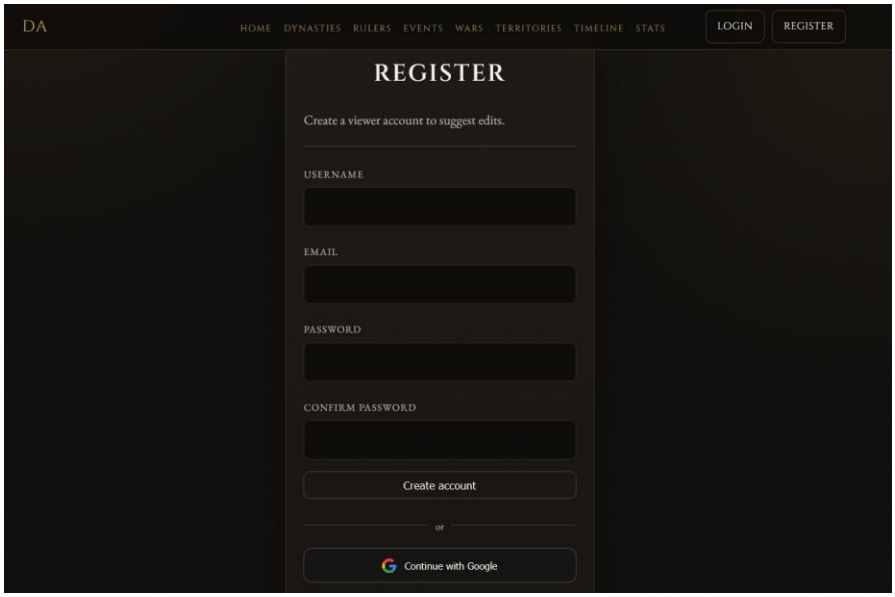
24. **Add / Edit Event Screen** — Form to create or update an event including name, type, dates, location, description, outcome, dynasty, image, and linked persons.
25. **Edit Requests Screen** — Admin view of all pending edit suggestions submitted by viewers. Admin can approve or decline each one.
26. **Backup Screen** — Triggers a database backup (pg_dump), uploads it to Firebase Realtime Database and Supabase Storage, and lists existing backups.
27. **Restore Screen** — Lists available backups from Firebase/Supabase; admin selects one, confirms with password, and the database is restored from that backup.
28. **Clear Database Screen** — Destructive action screen requiring the admin to type a confirmation phrase and enter their password before wiping all data.

Screenshots:

Login Screen



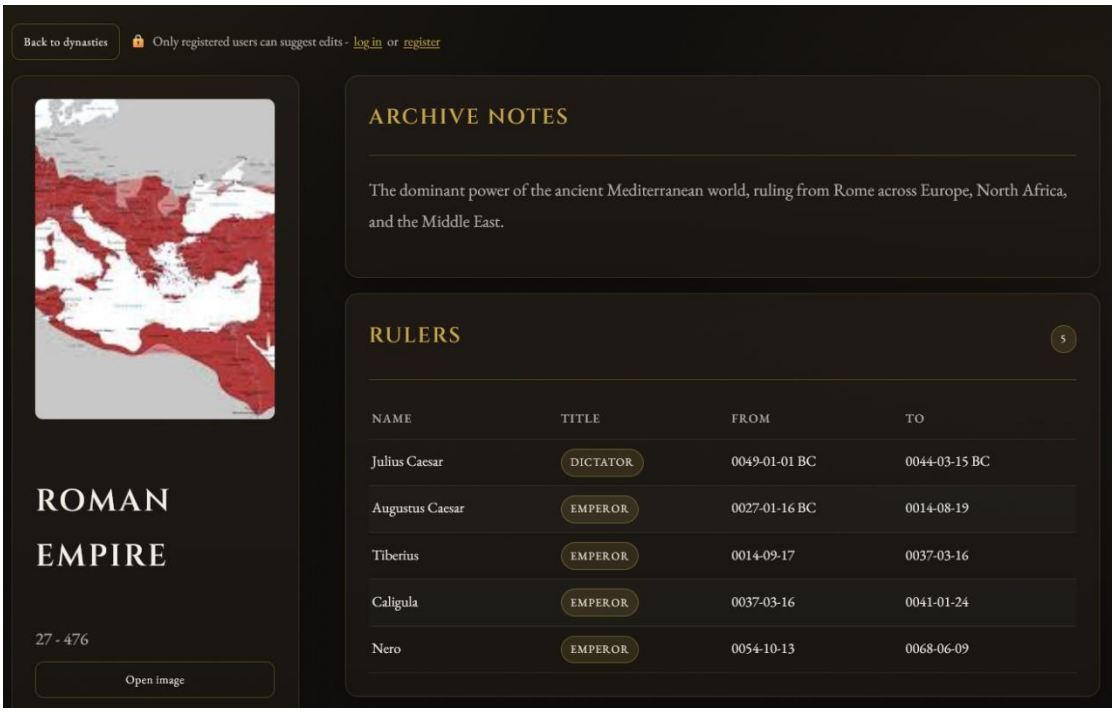
Register Screen



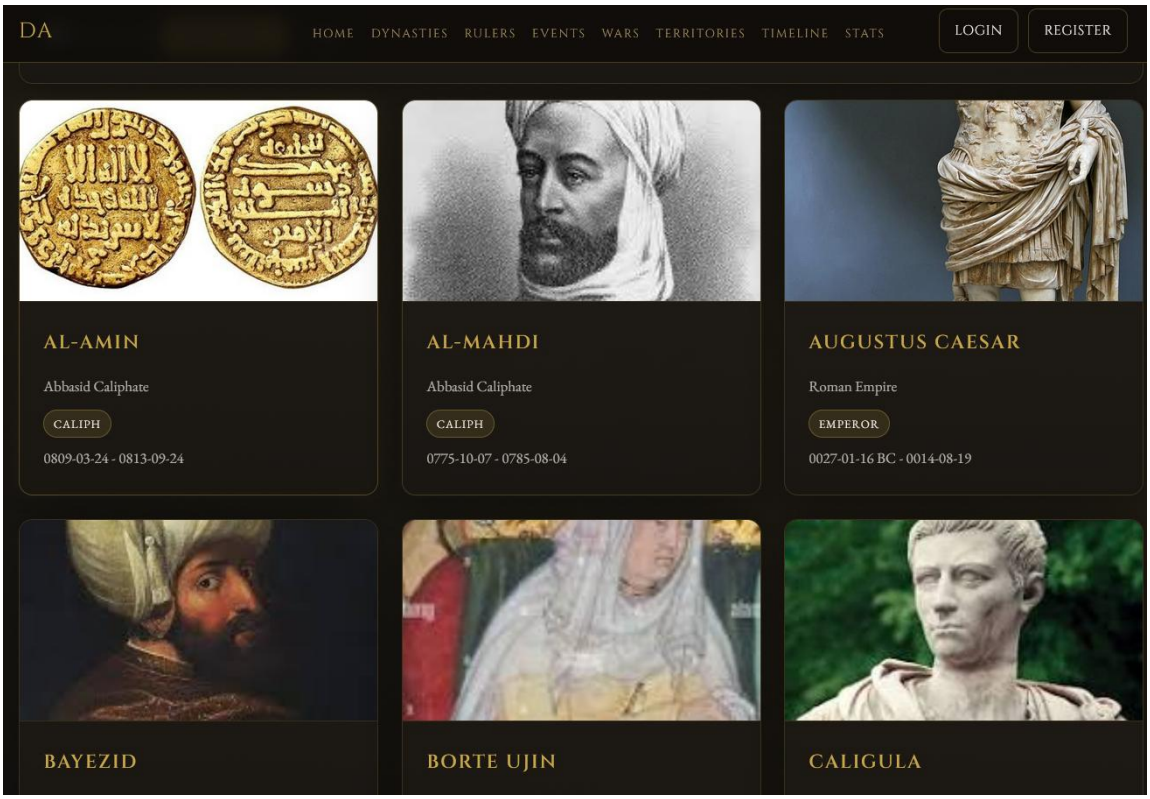
Home Screen

Dynasties List Screen

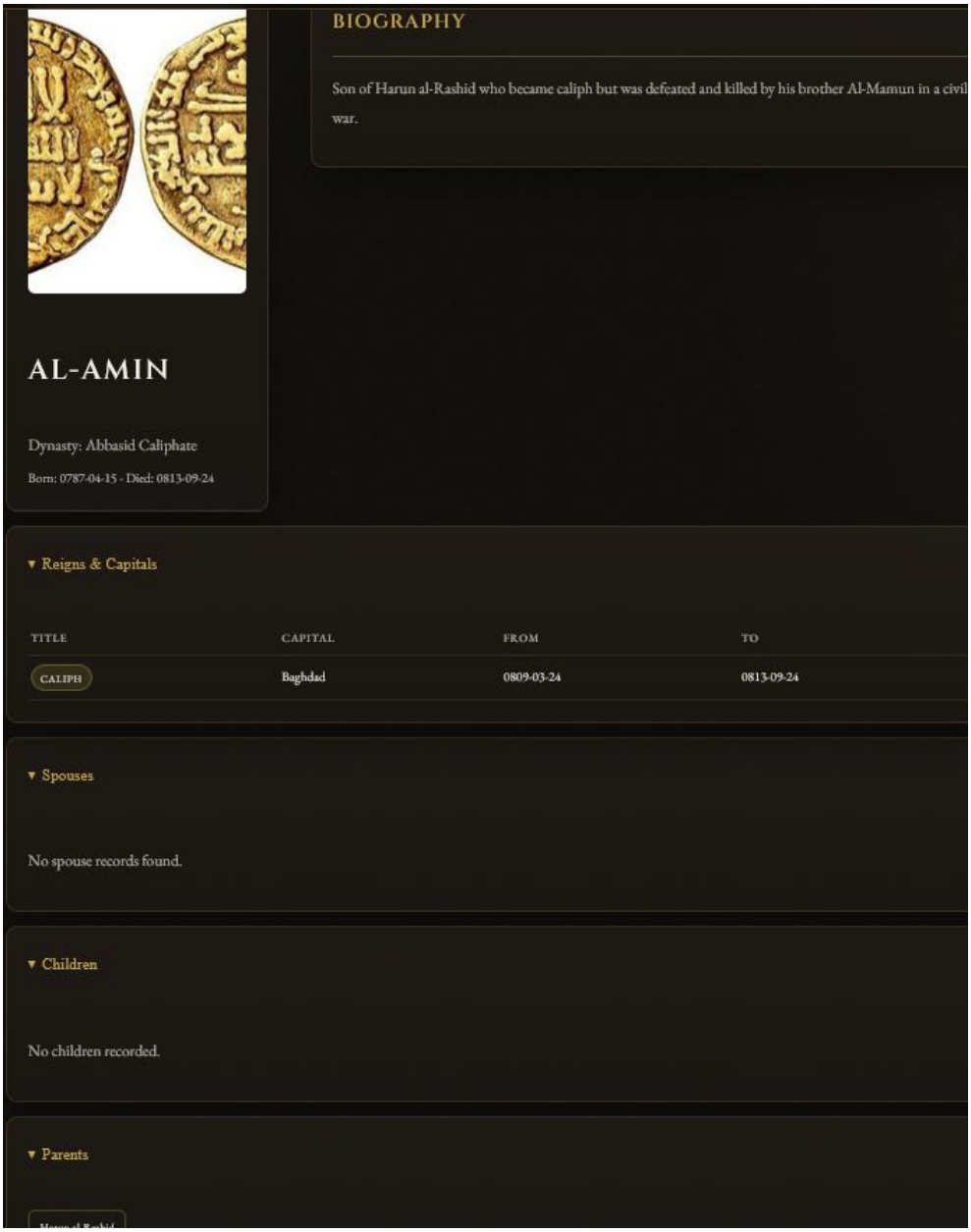
Dynasty Detail Screen



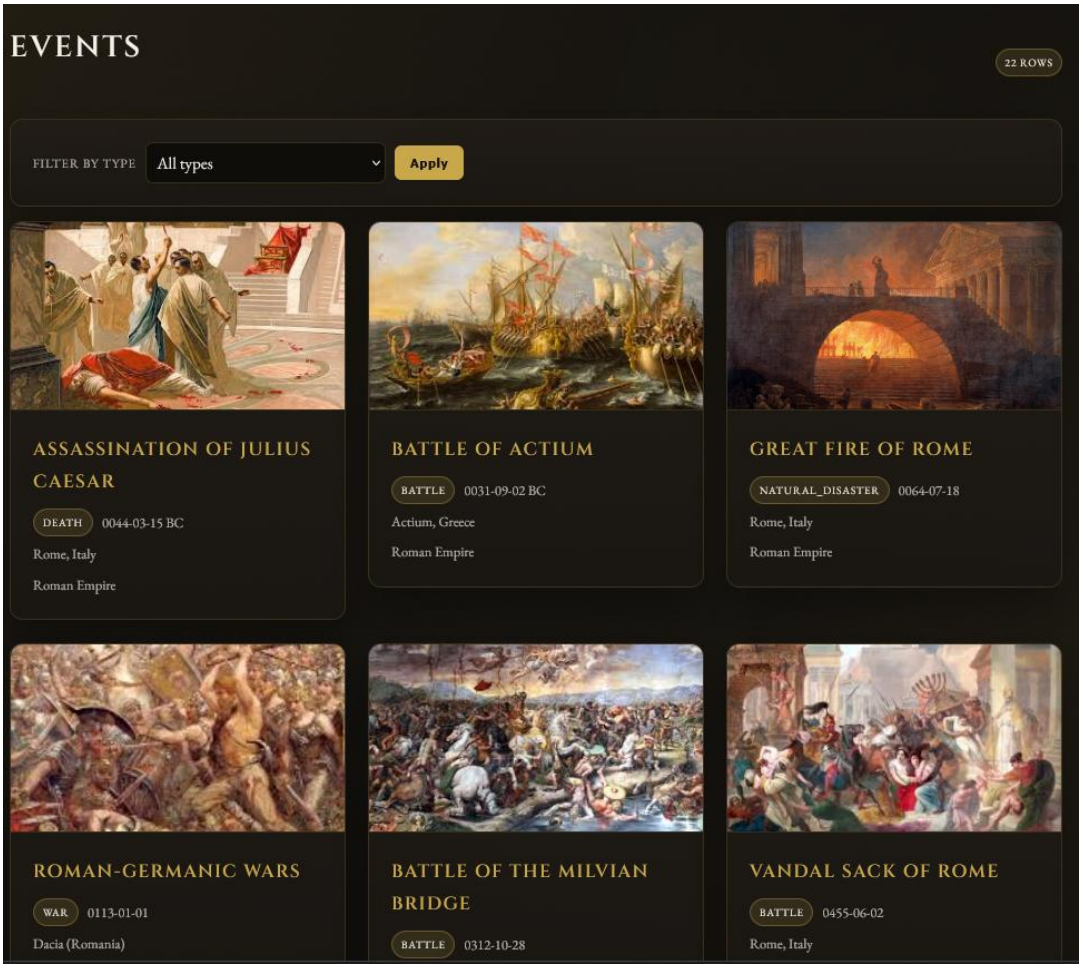
Rulers List Screen



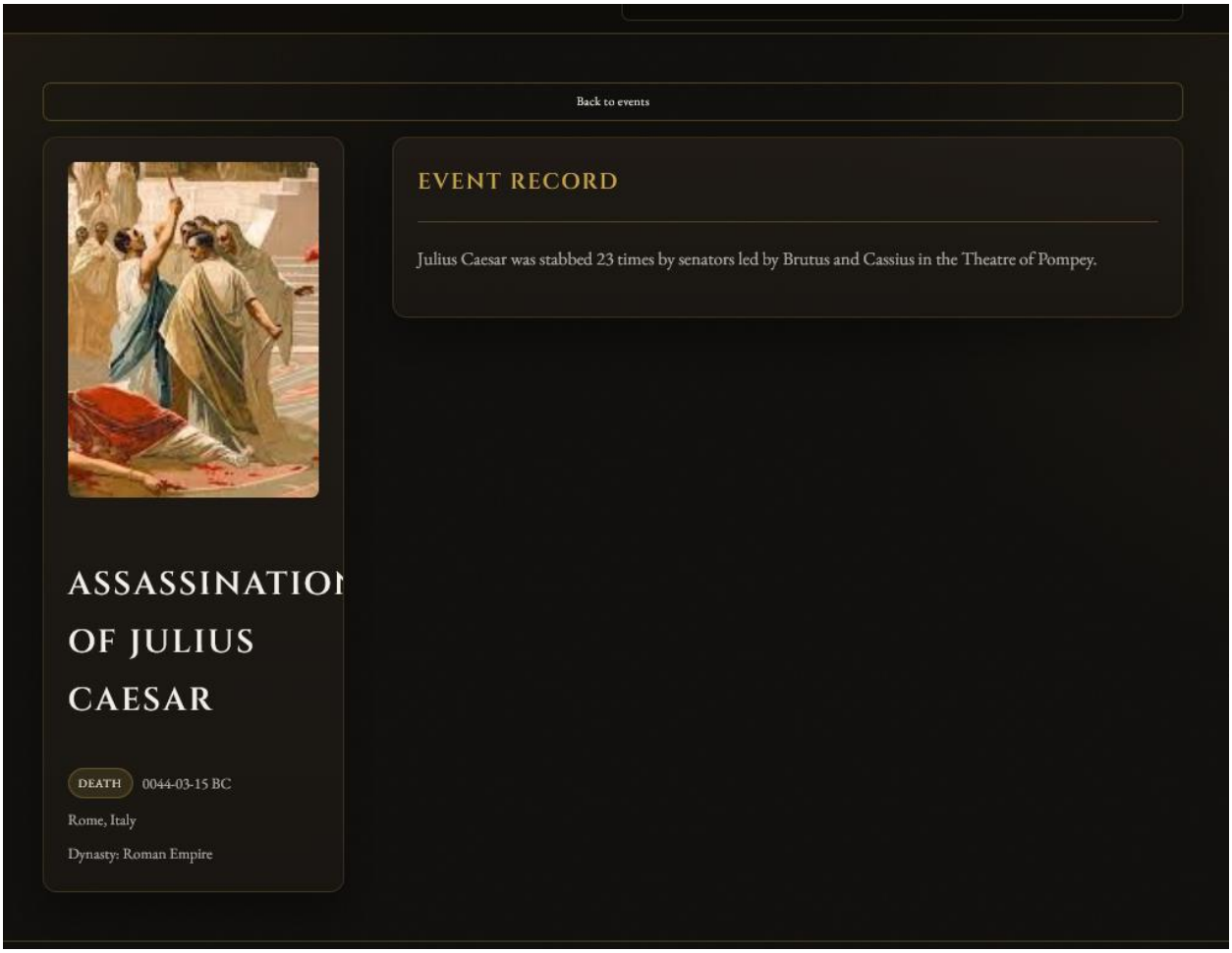
Ruler Detail Screen



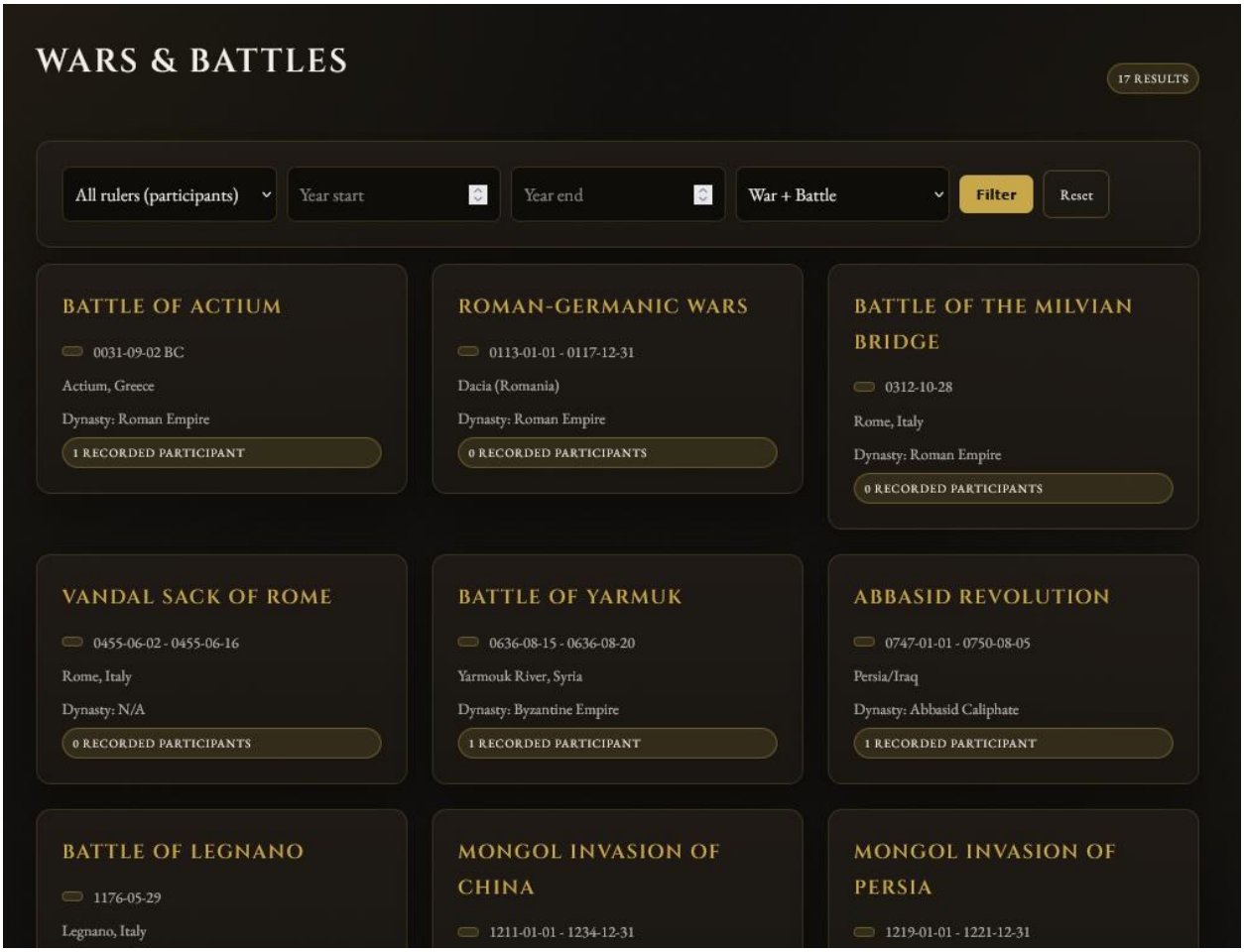
Events List Screen



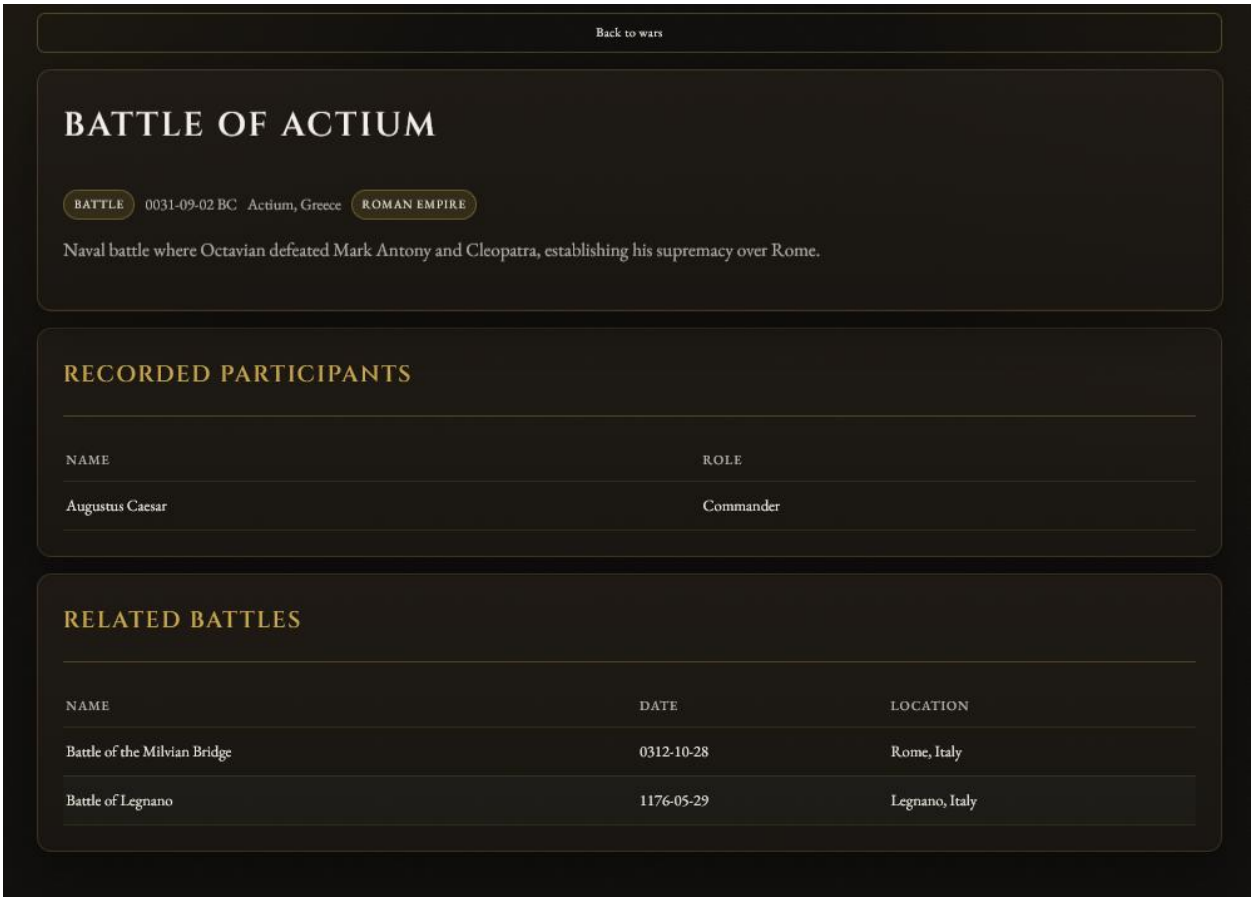
Event Detail Screen



Wars & Battles Screen

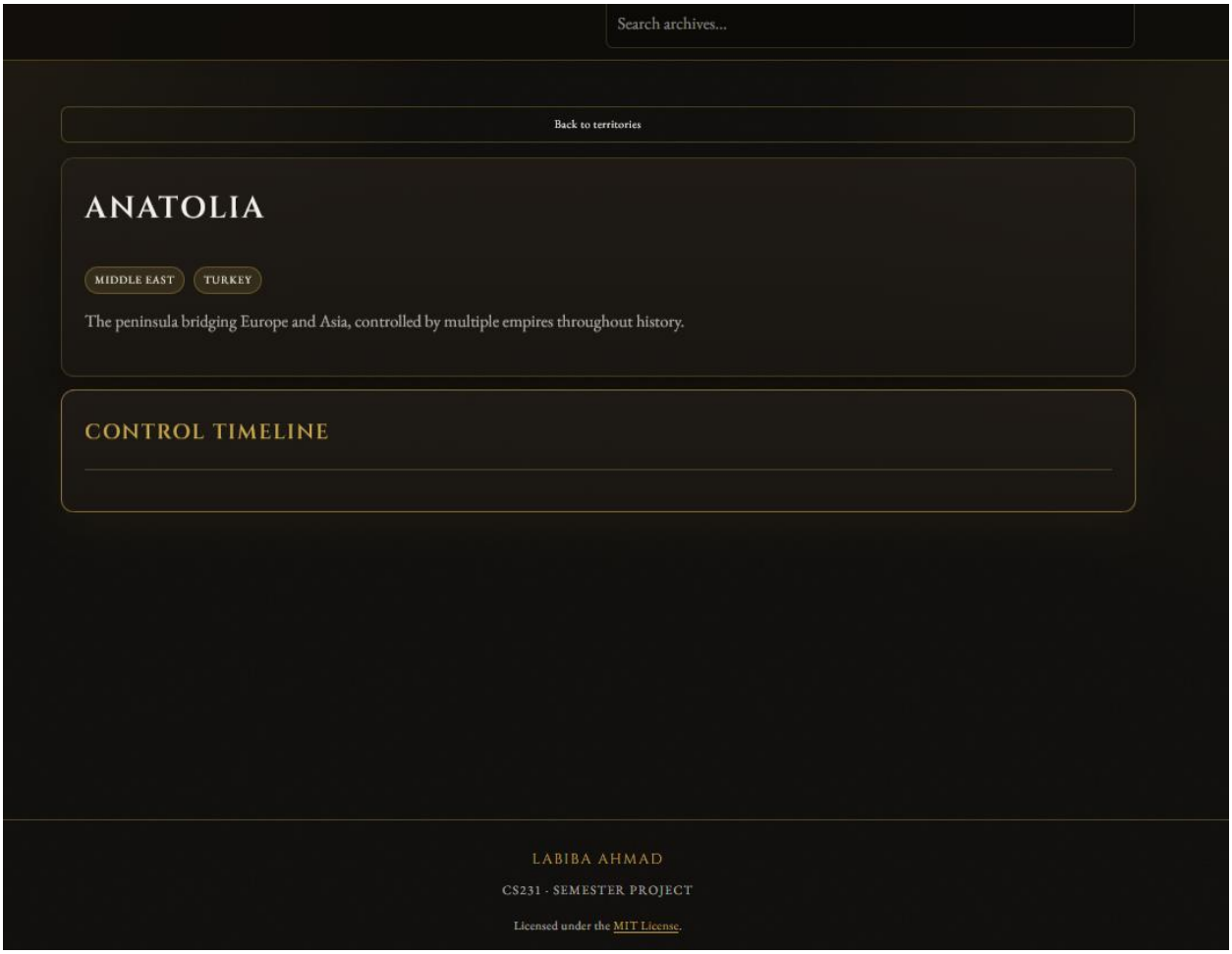


War/Battle Detail Screen

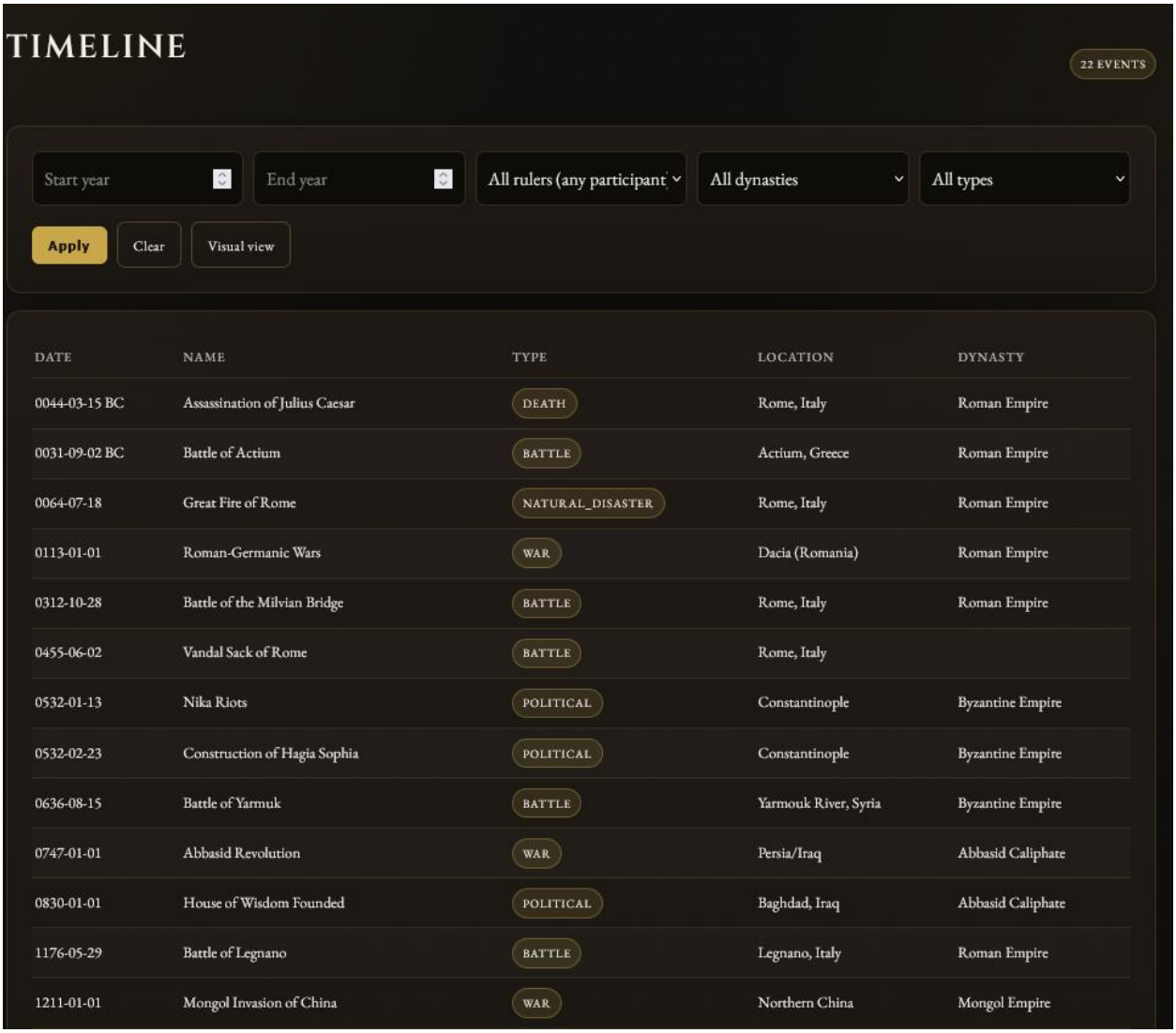


Territories List Screen

Territory Detail Screen

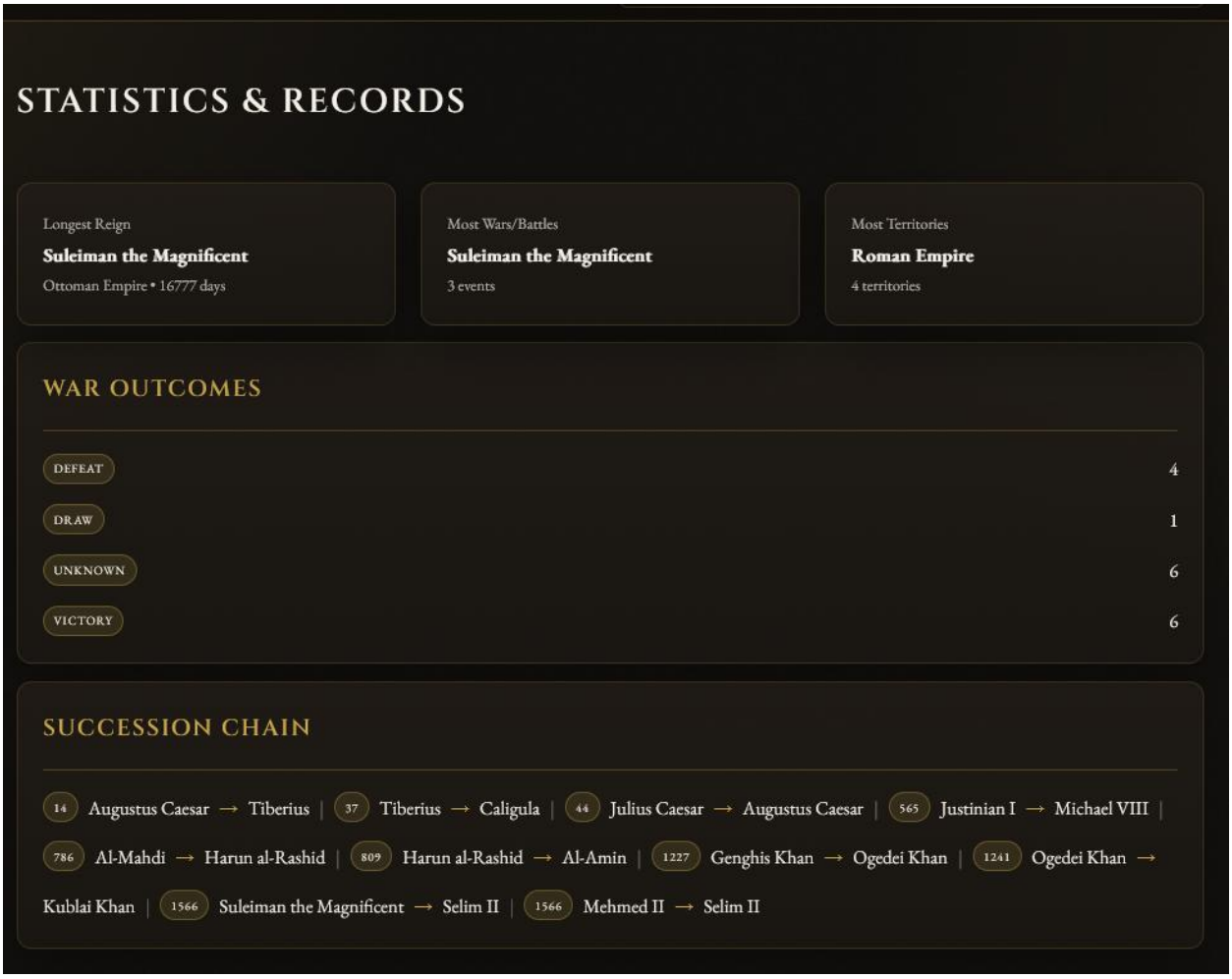


Timeline Screen

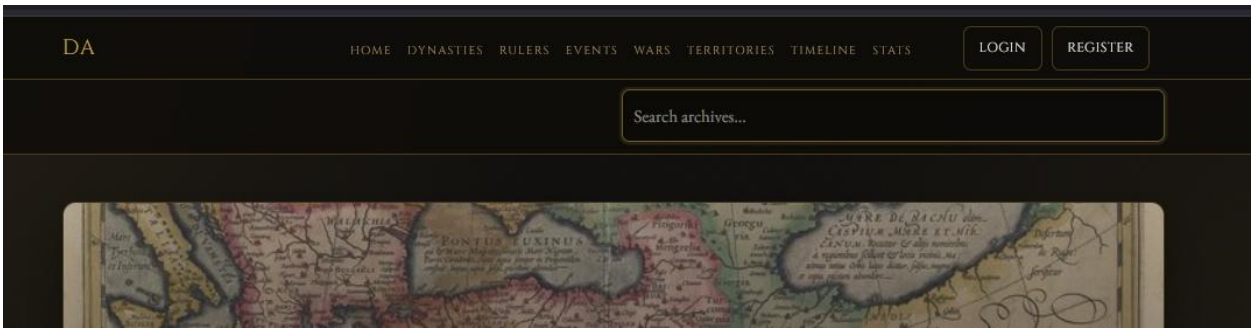


DATE	NAME	TYPE	LOCATION	DYNASTY
0044-03-15 BC	Assassination of Julius Caesar	DEATH	Rome, Italy	Roman Empire
0031-09-02 BC	Battle of Actium	BATTLE	Actium, Greece	Roman Empire
0064-07-18	Great Fire of Rome	NATURAL_DISASTER	Rome, Italy	Roman Empire
0113-01-01	Roman-Germanic Wars	WAR	Dacia (Romania)	Roman Empire
0312-10-28	Battle of the Milvian Bridge	BATTLE	Rome, Italy	Roman Empire
0455-06-02	Vandal Sack of Rome	BATTLE	Rome, Italy	
0532-01-13	Nika Riots	POLITICAL	Constantinople	Byzantine Empire
0532-02-23	Construction of Hagia Sophia	POLITICAL	Constantinople	Byzantine Empire
0636-08-15	Battle of Yarmuk	BATTLE	Yarmouk River, Syria	Byzantine Empire
0747-01-01	Abbasid Revolution	WAR	Persia/Iraq	Abbasid Caliphate
0830-01-01	House of Wisdom Founded	POLITICAL	Baghdad, Iraq	Abbasid Caliphate
1176-05-29	Battle of Legnano	BATTLE	Legnano, Italy	Roman Empire
1211-01-01	Mongol Invasion of China	WAR	Northern China	Mongol Empire

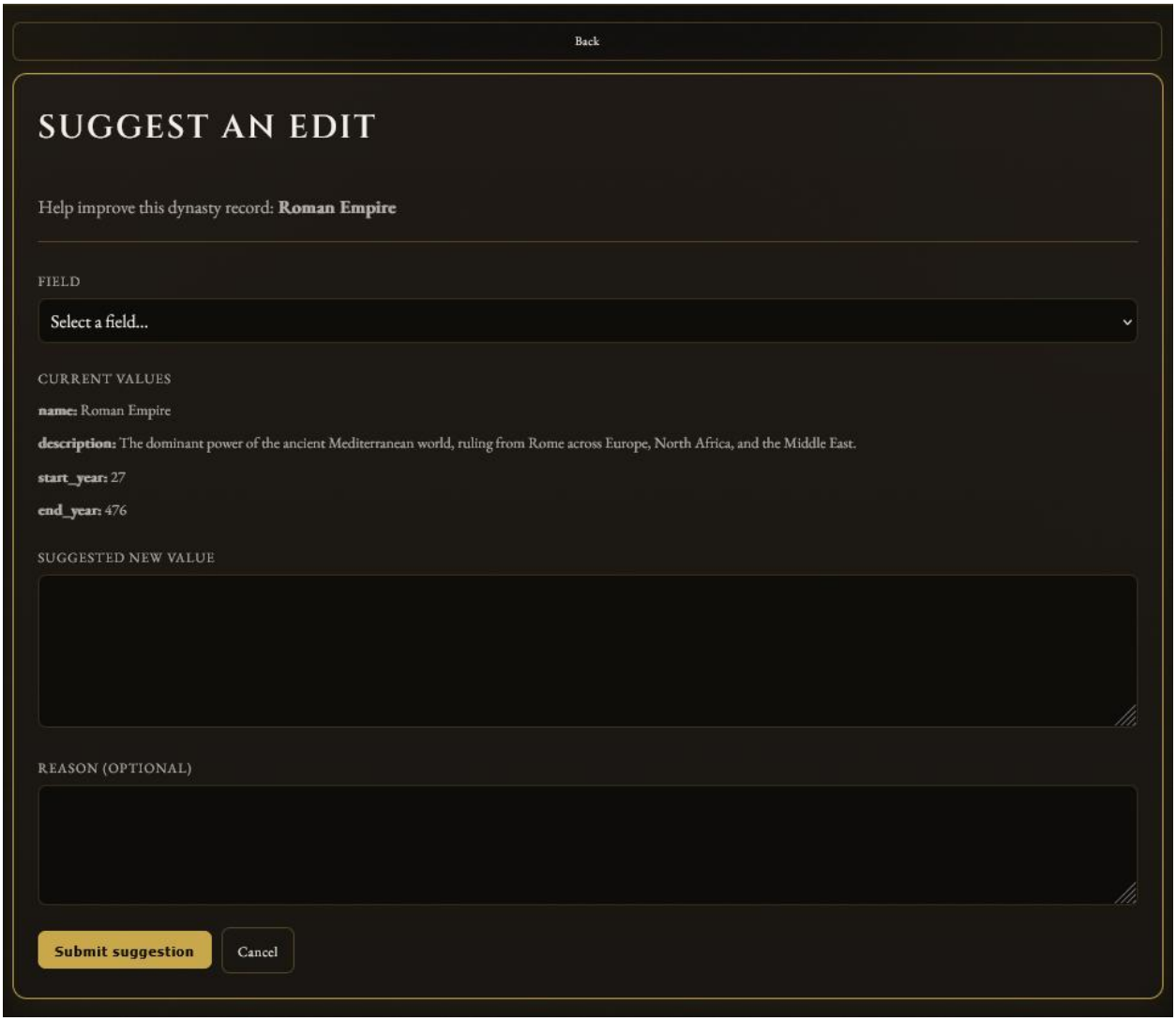
Stats Screen



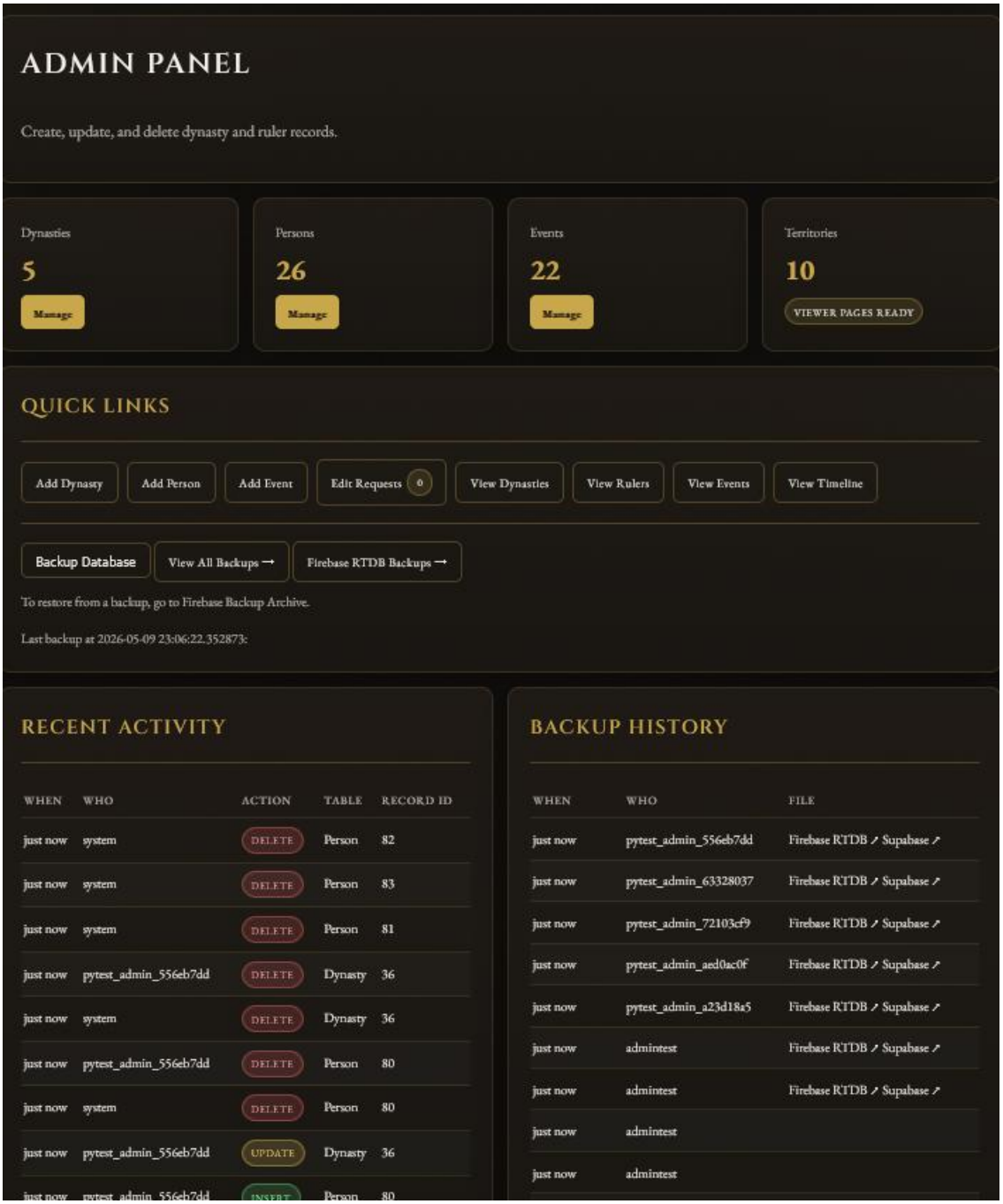
Search Screen



Suggest Edit Screen

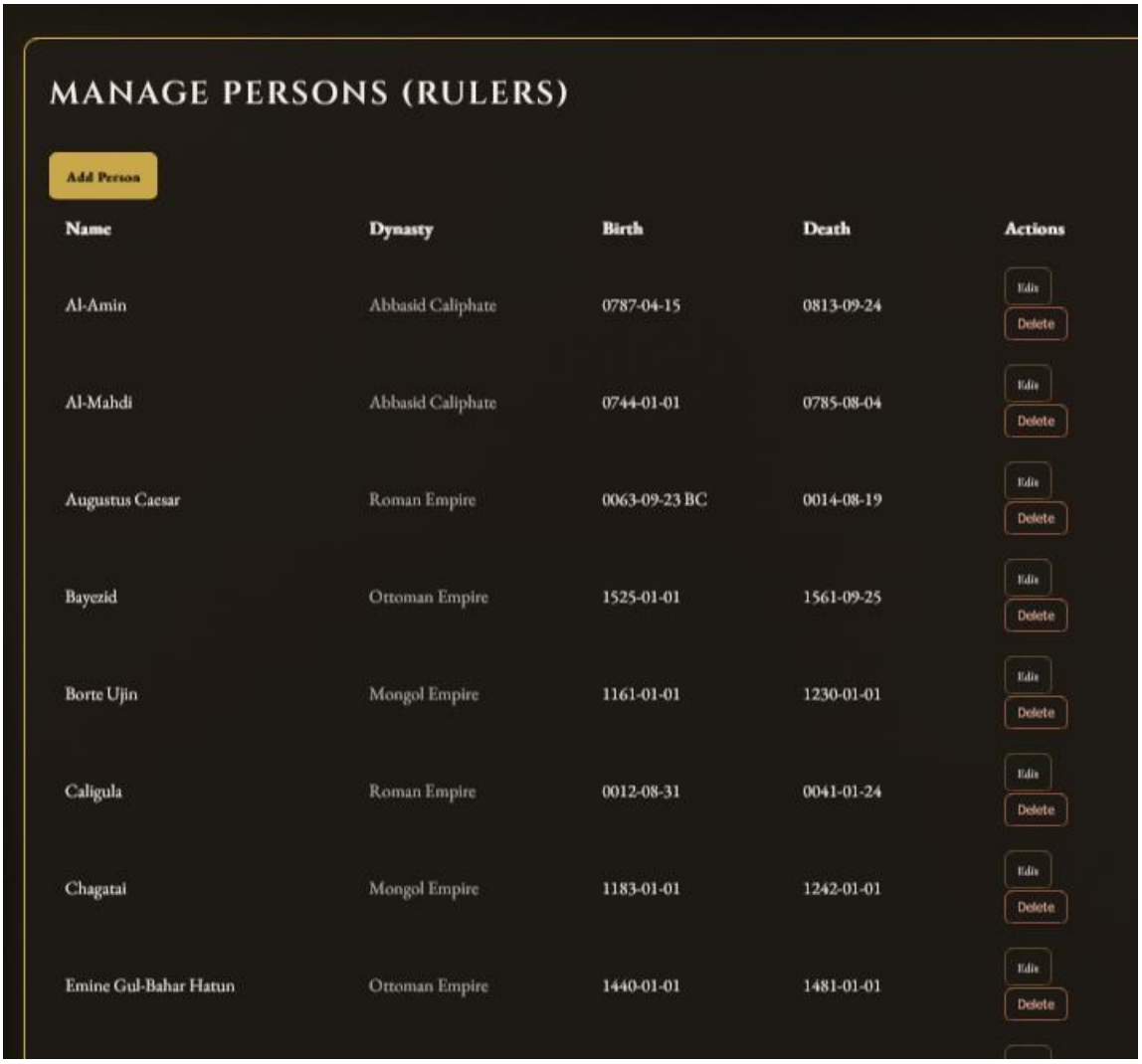


Admin Panel Screen

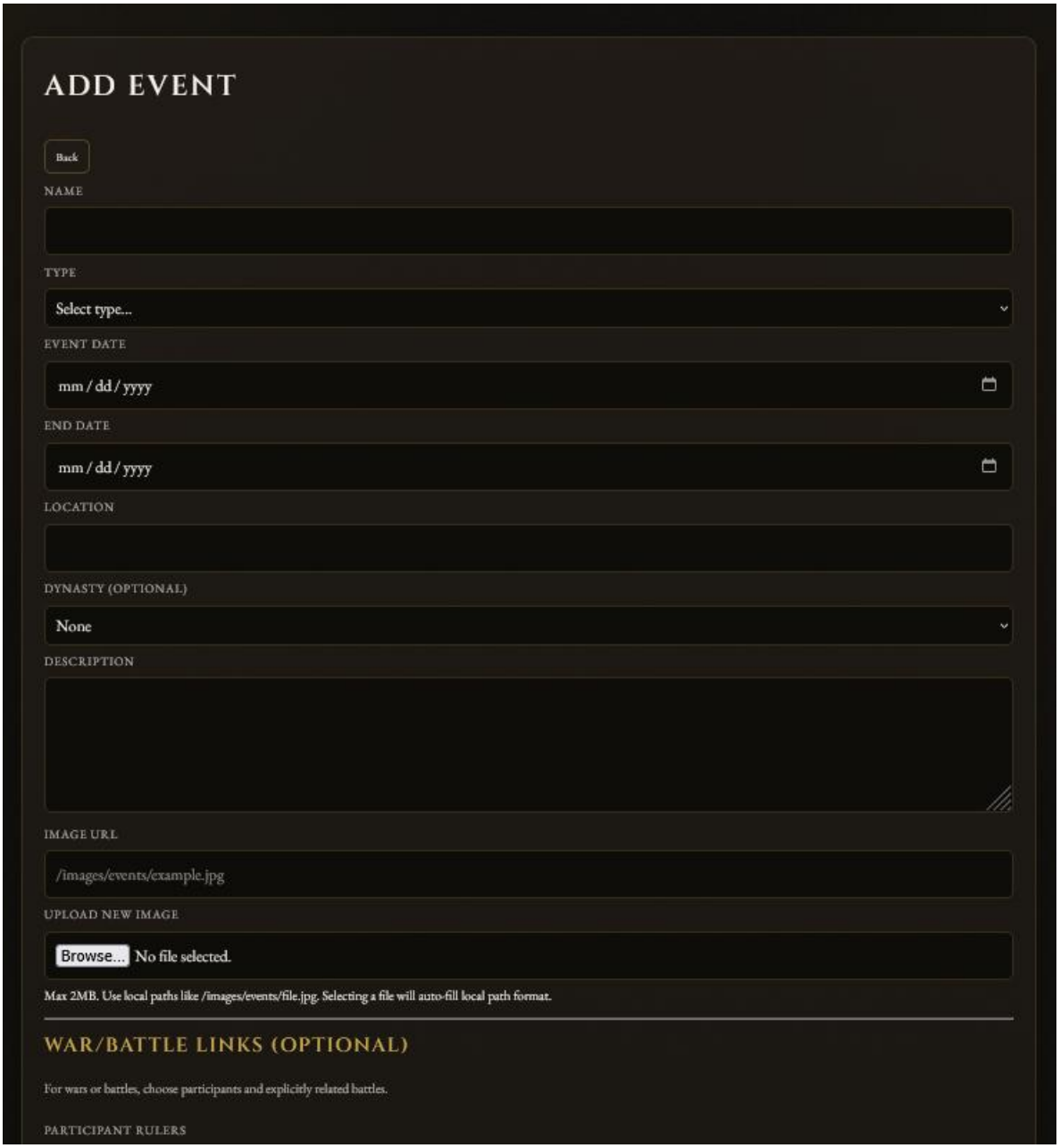


Admin Dynasties List Screen

Admin Persons List Screen

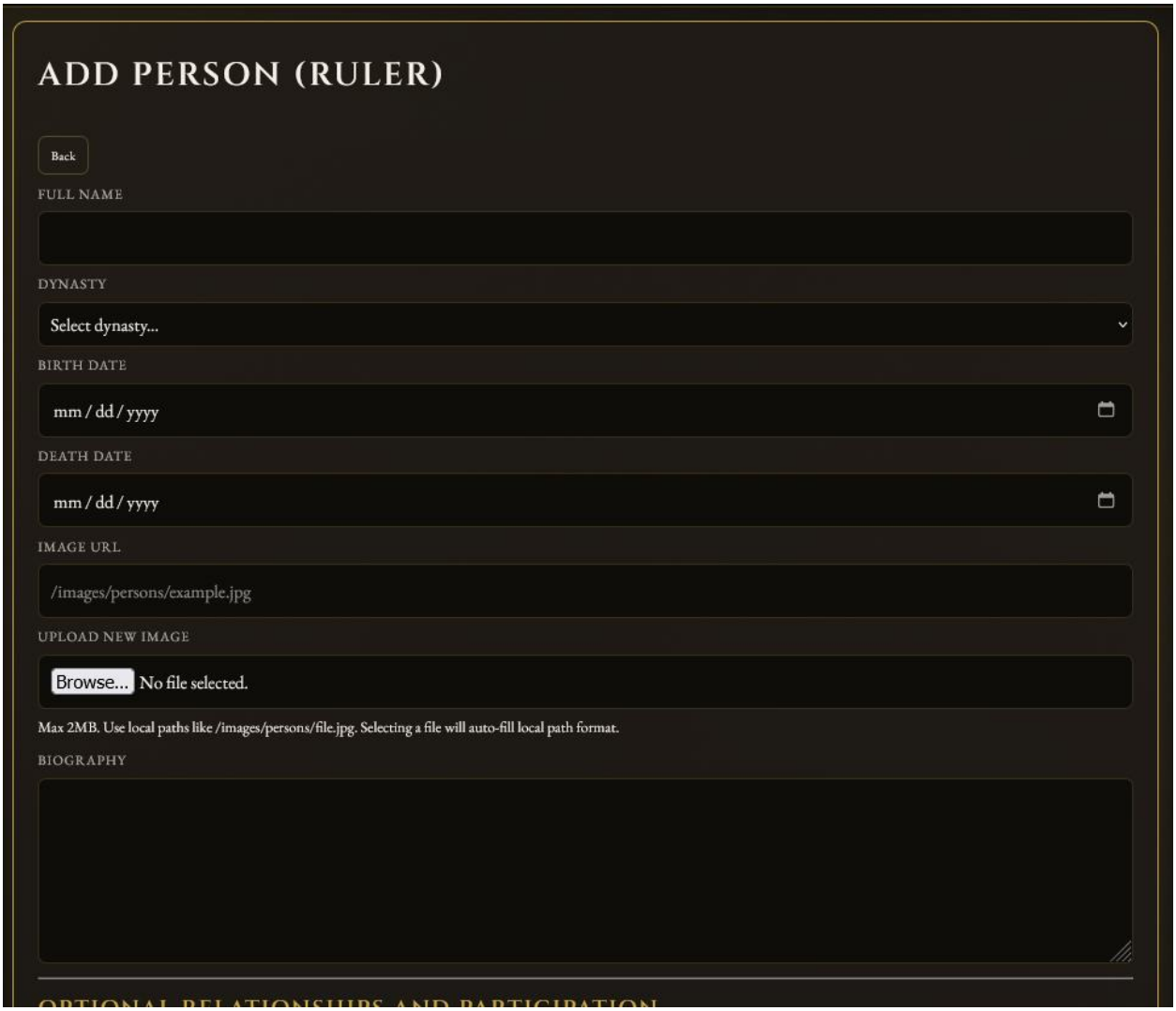


Admin Events List Screen



Admin Dynasty Screen

Admin Person Screen



Admin Dashboard - 2

RECENT ACTIVITY

WHEN	WHO	ACTION	TABLE	RECORD ID
just now	system	DELETE	Person	82
just now	system	DELETE	Person	83
just now	system	DELETE	Person	81
just now	pytest_admin_556eb7dd	DELETE	Dynasty	36
just now	system	DELETE	Dynasty	36
just now	pytest_admin_556eb7dd	DELETE	Person	80
just now	system	DELETE	Person	80
just now	pytest_admin_556eb7dd	UPDATE	Dynasty	36
just now	pytest_admin_556eb7dd	INSERT	Person	80
just now	pytest_admin_556eb7dd	INSERT	Dynasty	36

BACKUP HISTORY

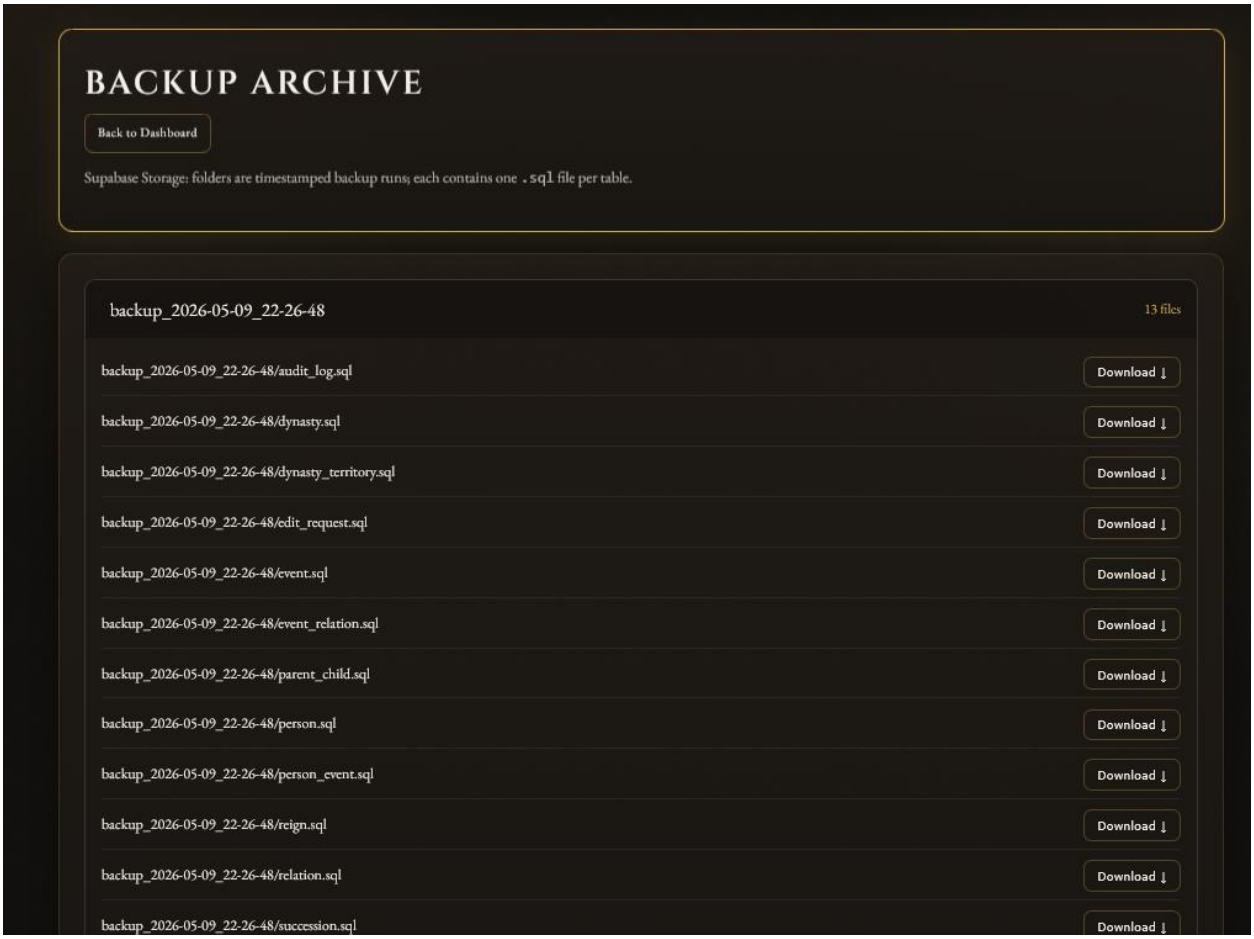
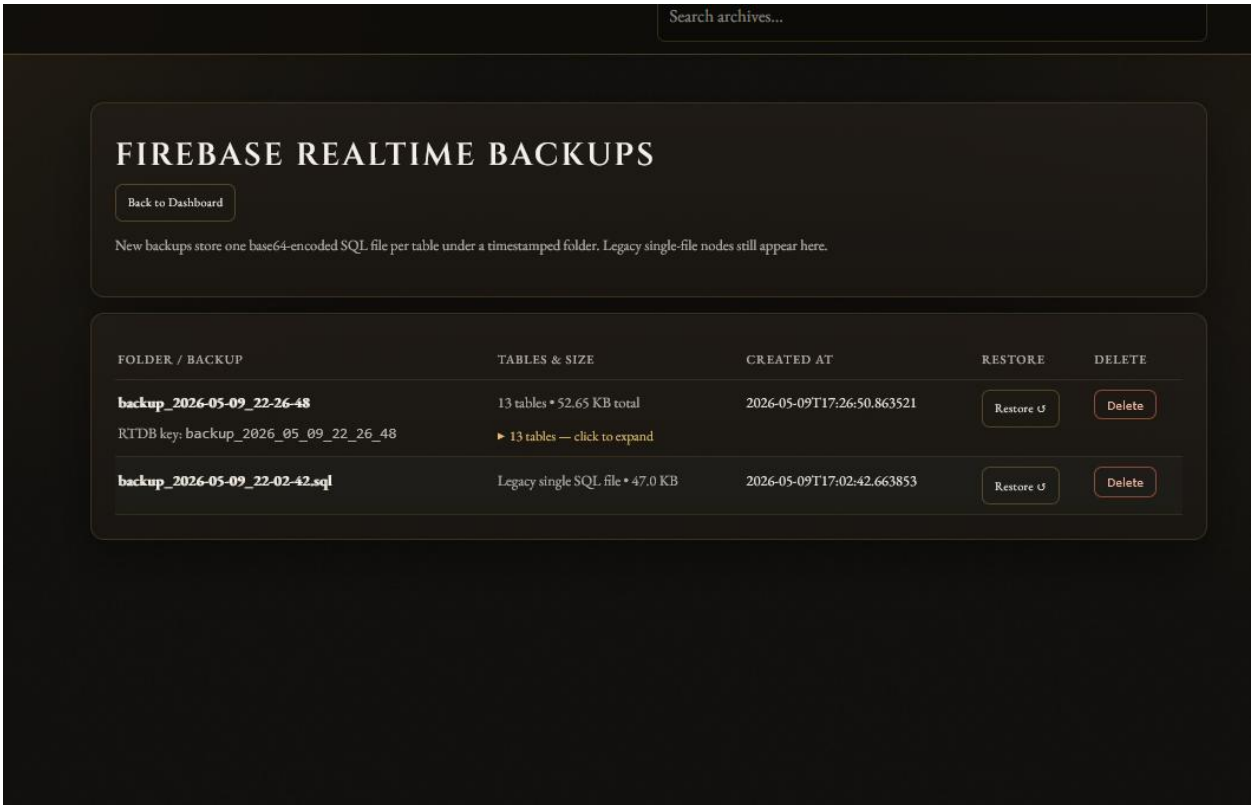
WHEN	WHO	FILE
just now	pytest_admin_556eb7dd	Firebase RTDB ↗ Supabase ↗
just now	pytest_admin_63328037	Firebase RTDB ↗ Supabase ↗
just now	pytest_admin_72103cf9	Firebase RTDB ↗ Supabase ↗
just now	pytest_admin_aed0ac0f	Firebase RTDB ↗ Supabase ↗
just now	pytest_admin_a23d18a5	Firebase RTDB ↗ Supabase ↗
just now	admintest	Firebase RTDB ↗ Supabase ↗
just now	admintest	Firebase RTDB ↗ Supabase ↗
just now	admintest	
just now	admintest	
7 minutes ago	admintest	Firebase RTDB ↗ Supabase ↗

DANGER ZONE

Permanently deletes all records. Cannot be undone.

Clear Database

Backup Screen



Restore Screen